

SELinux Assignment - Questions and Answers

Question 1: Check the current SELinux status on your Linux server using the `getenforce` and `sestatus` commands, and write down which mode is currently active.

Answer:

Step 1: `getenforce`

Step 2: `sestatus`

Example Output:

Enforcing

Current SELinux Mode: Enforcing

Question 2: Switch SELinux to Enforcing mode using `setenforce` and verify the change with `getenforce`.

Answer:

`sudo setenforce 1`

`getenforce`

Expected Output:

Enforcing

Question 3: Simulate an SELinux access denial by creating a new directory and setting an incorrect context.

Answer:

`sudo mkdir -p /tmp/testapp`

`echo "SELinux Test Page" | sudo tee /tmp/testapp/index.html`

`sudo chcon -R -t user_home_t /tmp/testapp`

`ls -Zd /tmp/testapp`

Try accessing the file through the web server.

Check logs:

`sudo tail -20 /var/log/audit/audit.log`

or

`sudo tail -20 /var/log/messages`

Example Error:

SELinux is preventing httpd from accessing /tmp/testapp/index.html

Question 4: Use `chcon` to assign the correct SELinux context and confirm the issue is resolved.

Answer:

`sudo chcon -R -t httpd_sys_content_t /tmp/testapp`

`ls -Zd /tmp/testapp`

`sudo systemctl restart httpd`

Reload the webpage in the browser.

The page should now open successfully.

Question 5: Document the steps taken to identify and resolve the SELinux access denial.

Answer:

1. Checked SELinux status using `getenforce` and `sestatus`.
2. Verified SELinux was running in Enforcing mode.
3. Created `/tmp/testapp` and assigned an incorrect SELinux context (`user_home_t`).
4. Attempted to access the file through the web server and received an access denial.
5. Checked audit logs to identify the SELinux denial message.
6. Changed the context to `httpd_sys_content_t` using `chcon`.
7. Restarted the web server and verified successful access.

Example Log Entry:

`type=AVC msg=audit(...) : denied { read } for pid=1234 comm="httpd" name="index.html"`

Brief Explanation:

SELinux uses security contexts to determine which processes can access specific files and directories. If a file has the wrong context, SELinux blocks access even when traditional Linux permissions allow it. Assigning the correct context enables secure access while maintaining system security.